Kumkum Mishra

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EDUCATION

BTech CSE	Sharda University, Uttar Pradesh	CGPA: **9.21/10**	July 2023 - July 2027
Intermediate(CBSE)	Govt. Girls Senior Sec. School, Delhi	CGPA: **9.14/10**	April 2022 - April 2023
High School(CBSE)	Rajkiye Sarvodaya Kanya Vidhyalaya, Delhi	CGPA: **9.08/10**	April 2020 - April 2021

PROFESSIONAL SUMMARY

Machine Learning Researcher focused on investigating challenges in **machine learning, large language models, and computer vision**. Experienced in formulating experimental approaches, evaluating competing methods, and analyzing model behavior to understand limitations and identify potential improvements. My work explores **automated program repair, reliable and efficient LLMs, and carbon-aware prompt engineering**, with an emphasis on empirical evidence, rigorous evaluation, and advancing practical understanding of intelligent systems.

SKILLS

Languages: C++, Python, SQL
Core Skills: Python Scripting, Data Processing, Python-based Machine Learning Pipelines, REST API Integration for ML Services
AI/ML: Deep Learning, Computer Vision, Transformer-based Models, Large Language Models (LLMs), Retrieval Augmented Generation(RAG), Langchain, Model Evaluation & Experimental Analysis
Frameworks & Libraries: FastAPI, Flask, Streamlit, Scikit-Learn, Pandas, NumPy, Matplotlib
Databases & Tools: Git, GitHub, MySQL, SQLite
Soft Skills: Clear Technical Communication, Team Collaboration, Deadline-Driven Execution

WORK EXPERIENCE

Freelancing - Upwork September 2025 – September 2026 (Remote)
Freelance Machine Learning Engineer – LLM Systems | Deep Learning

- Developed an end-to-end **automated program repair (APR) pipeline** using CodeT5 and 10+ transformer models, generating **4,700+ candidate patches across 79 Python and Java bugs** evaluated using program-repair benchmarks.
- Conducted systematic experiments on code-generation models, including **OpenCoder-1.5B-Instruct**, across **MBPP, HumanEval, and BigCodeBench**, evaluating pass@1 performance, few-shot prompting strategies, and domain-shift behavior.
- Investigated **performance–efficiency trade-offs** across FP32, FP16, INT8, and INT4 configurations, measuring inference latency and energy consumption using **CodeCarbon**.
- Built modular **LLM inference and evaluation pipelines** using Hugging Face Transformers and vLLM, incorporating patch validation, output sanitization, and structured experiment logging.

Birla Institute of Technology and Science, Pilani (BITS Pilani) April 2025 – July 2025 (Remote)
Research Intern – **Machine Learning | Deep Learning | Semantic Segmentation | PyTorch**

- Evaluated semantic segmentation models on the Indian Driving Dataset (IDD) and its adverse-weather extension (IDD-AW), benchmarking performance across fog, rain, low-light, and snow conditions using mean Intersection over Union (mIoU).
- Implemented a four-phase experimental pipeline comparing **DeepLabV3+**, weather-based data augmentation, **HRNet with an OCR head**, and **SegFormer-B0**; recorded the highest reported adverse-weather **mIoU** of **10.97% in fog** and **9.53% in rain** with SegFormer-B0.
- Developed PyTorch data-loading and preprocessing pipelines, converted JSON annotations into segmentation masks, and documented model configurations, evaluation results, and prediction visualizations to support reproducible experiments.

PROJECT

CleanForge - AI Data Cleaning & Quality Platform January-2026 - February-2026
Tech Stack – **FastAPI | Next.js | Pandas | Groq API | Render | Vercel** GitHub

- Built a **full-stack AI data cleaning platform** for end-to-end **CSV profiling, scoring, and automated cleaning**, implementing pipelines for **duplicate removal, normalization, numeric coercion, and IQR-based outlier handling** across diverse real-world datasets.
- Designed a **composite data quality scoring system (45% completeness, 25% duplicates, 30% outliers)** and integrated **LLM-based semantic analysis** to detect column-level issues, deploying a **production-ready system** with FastAPI APIs and Next.js UI.

CarbonPrompt – Evaluating Prompt Engineering and Emissions in LLMs June - 2026 - August - 2026
Tech Stack – Python | Hugging Face Transformers | FLAN-T5-Small | CodeCarbon | Prompt Engineering GitHub

- Developed **CAPEF (Carbon-Aware Prompt Engineering Framework)** to profile LLM prompts, estimate Green Carbon Scores (GCS), and optimize prompts using a fine-tuned **FLAN-T5-Small** model trained on **10,000 prompt pairs**.
- Reduced mean prompt length by **16.3% across 200 prompts** (50.7 to 44.0 tokens), achieving a **0.94 similarity score** with FLAN-T5, compared with 0.77 for the local optimizer and 0.82 for the API optimizer.

ACHIEVEMENTS & HACKATHONS

- **Cyber Cup 5.0(Amity University): 2nd Runner-Up** for an AI carpooling system reducing traffic and carbon emissions.
- **SIH 2025: Internal level finalist** for building a sustainable AI carbon-aware prompting tool.
- **Recommendation letter** from **BITS Pilani.**

CERTIFICATES

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate October 2025 – October 2025
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional October 2025 – October 2025

PUBLICATIONS

- **Shah, S., Mishra, K., & Gola, D. K.** (2026). *A Multi-Task Benchmark for Prompt Engineering in Large Language Models: Performance, Efficiency and Reliability Trade-offs.* **International Journal of Innovative Science and Research Technology**, 11(4), April 2026. DOI: 10.38124/ijisrt/26apr1597. **[Published]**

- **Mishra, K., et al.** (2026). *Carbon Prompting: An Empirical Analysis of LLM-Based Requirement Classification.* International Conference on Multidisciplinary Perspectives in Advanced Computing and Technology (**IMPACT 2026**). **[Published]** (Link)

- **Mishra, K., et al.** *A Carbon-Aware Prompt Engineering Framework for Sustainable Large Language Model Inference.* **Engineering Management Journal. [Under Review]**